

Use separate answer scripts for each group of questions

Group - A (40 Marks)

Answer any eight questions:- (5x8=40)

1. Express $\begin{pmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \\ 5 & 6 & 7 \end{pmatrix}$ as the sum of a symmetric and a skew symmetric matrix.

2. Show that $\begin{vmatrix} 1 & a & a^2 \\ a^2 & 1 & a \\ a & a^2 & 1 \end{vmatrix}$ is a perfect square. Verify the result for $a = 3$.

3. Find characteristic equation and eigenvalues of the

$$\begin{pmatrix} 1 & -1 & 0 \\ 1 & 2 & -1 \\ 3 & 2 & -2 \end{pmatrix}$$

matrix

4. State Cayley Hamilton theorem. Use Cayley Hamilton theorem

$$\text{to find } A^{50}, \text{ where } A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

5. Show that

$$\begin{vmatrix} 1 & \cos x - \sin x & \cos x + \sin x \\ 1 & \cos y - \sin y & \cos y + \sin y \\ 1 & \cos z - \sin z & \cos z + \sin z \end{vmatrix} = 2 \begin{vmatrix} 1 & \cos x & \sin x \\ 1 & \cos y & \sin y \\ 1 & \cos z & \sin z \end{vmatrix}$$

6. If a, b, c are real numbers, prove that

$$a^2 + b^2 + c^2 \geq ab + bc + ca.$$

7. Apply elementary row operation to reduce the following matrix

$$\begin{pmatrix} 2 & 0 & 4 & 2 \\ 3 & 2 & 6 & 5 \\ 5 & 2 & 10 & 7 \\ 0 & 3 & 2 & 5 \end{pmatrix}$$

to a row echelon matrix

8. Solve the following system of equations by Cramer's method $x+2y-3z=1$; $2x-y+z=4$; $x+3y=5$.

10. Apply matrix method to show that $x=3, y=1, z=2$ is solution of the system of equations

$$2x + 3y + z = 11, \quad x + y + z = 6 \text{ and } 5x - y + 10z = 34.$$

12. Prove that the intersection of two subspaces of a vector space V over a field F is a subspace of V .

14. Prove that the set of vectors $\{(1,2,2), (2,1,2), (2,2,1)\}$ is linearly independent in $\mathbb{R}^3$.

9. Find the inverse of $A = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 4 & 4 \\ 3 & 3 & 7 \end{pmatrix}$.

11. If α, β, γ are roots of the equation $x^3 + px^2 + qx + r = 0$, find the value of $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2$ in terms of p, q and r .

13. Find $\text{mod } z$ and $\text{arg } z$ and express z in polar form where $z = -1+i$. $\frac{1}{\sqrt{2}}$ $(1+i\sqrt{3})$

15. Find the rank of $\begin{pmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 8 & 6 \\ 3 & 6 & 6 & 3 \end{pmatrix}$

Group - B(40 Marks)

1. Answer the following questions: 1X5 = 5

(a) Under what condition the following differential equation $(2ax + by + g)dx + (2fy + cx + h)dy = 0$ is it exact?

(b) Find the integrating factor of the differential equation:

$$x \cos x \frac{dy}{dx} + (x \sin x + \cos x)y = 1.$$

(c) Find the complementary function of the differential equation: $\frac{d^3y}{dx^3} + 2\frac{d^2y}{dx^2} + \frac{dy}{dx} = e^{2x}$.

(d) Find the complementary function of the differential equation: $\frac{d^2y}{dx^2} - 4y = \sin 2x$.

(e) If $y_1 = e^x$ and $y_2 = e^{2x}$, then find the Wronskian $W(y_1, y_2)$ of y_1 and y_2 .

3. Solve ANY TWO differential equations:

5X2=10

(a) $\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} + 3\frac{dy}{dx} - y = xe^x + e^x$.

(b) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 5\cos x$, given that $y = \frac{dy}{dx} = 0$, when $x = 0$.

(c) $x^2 \frac{d^2y}{dx^2} + 2x \frac{dy}{dx} - y = 3e^x \cos \log(x)$.

(d) $x^2 \frac{d^2y}{dx^2} + 5x \frac{dy}{dx} + 4y = x^4$.

16. When is an $n \times n$ matrix called a row reduced Echelon Matrix? Give an example of it.

2. Solve ANY THREE differential equations: 5X3=15

(a) $(y^2 e^{xy^2} + 4x^3)dx + (2xye^{xy^2} - 3y^2)dy = 0$.

(b) $3e^x \tan x \tan y dx + (1 - e^x) \sec^2 y dy = 0$.

(c) $(y^4 - 2x^3y)dx + (x^4 - 2xy^3)dy = 0$.

(d) $\frac{dy}{dx} = \frac{y-x+1}{y+x+5}$.

(e) $x(1 - x^2) \frac{dy}{dx} + (2x^2 - 1)y = ax^3$, a being a fixed constant.

(f) $\frac{dy}{dx} + \frac{y}{x} = \frac{y}{x^2}(y)^2$.

4. Answer ANY ONE question: 6+4=10

(a) (i) Using the Method of variation of parameters, find the general solution of the differential equation:

$$\frac{d^2y}{dx^2} + a^2y = \tan ax.$$

(ii) Find the eigenvalues λ_n and the corresponding eigenfunction

$$y_n \text{ for the differential equation } \frac{d^2y}{dx^2} + \lambda y = 0 \quad (\lambda > 0)$$

satisfying the boundary conditions $y(0) = 0$ and $y(\pi) = 0$.

(b) (i) Using the Method of Undetermined Coefficients, find the general solution for the differential equation:

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + y = x^2 - 2x + 2.$$

(ii) Find the eigenvalues λ and the corresponding eigenfunction for the differential equation $\frac{d^2y}{dx^2} + \lambda y = 0$ ($\lambda > 0$) satisfying the boundary conditions

$$y(0) = 0 \text{ and } \frac{dy}{dx} = 0 \text{ at } x = \pi.$$